

PHENOVA — MASTER PROGRAM SPECIFICATION

Expanded Competitive Edition — Five-Product Capability Integration

Base specification retained; this document adds a competitive capability layer derived from current feature sets of OpusClip, Descript, CapCut, Videoleap, and Adobe Premiere.

Status: Amendment / expansion of the current PHENOVA production specification. The existing binding rules remain in force. This document does not replace the existing security, quota, architecture, testing, IFEC, or Definition-of-Done requirements.

Source basis: the supplied PHENOVA Master Program Specification and current PHENOVA implementation blueprint, plus current official product documentation for the five reference editors. Where a limitation is not explicitly documented by a vendor, it is described as a design opportunity rather than a confirmed defect.

1. Competitive Reference Set

Reference	Current strength	Capability to study
OpusClip / AI Producer	AI-first automatic editing	Agentic first-draft creation, edit explanations, text + timeline control, captions, B-roll, motion, music
Descript	Transcript/document-style editing	Transcript editing, AI co-editor, filler/silence cleanup, audio enhancement, regeneration, captions
CapCut	Broad mobile/creator toolkit	Auto Cut, beat/speech-aware editing, Smart Search, Auto Reframe, captions, templates, AI generation
Videoleap	Creative mobile AI editing	Creative effects, compositing, visual manipulation, AI-assisted transformations and mobile-first workflow
Adobe Premiere	Professional editing depth	Professional timeline, media intelligence, AI Assistant, object masking, generative extension, generative

2. Competitive Capability Inventory

- **Autonomous editing:** generate a complete first draft from raw footage; select clips, remove dead air/fillers, build pacing, add captions, B-roll, music, SFX, motion, layouts, and publish-ready formatting.
- **Natural-language editing:** allow users to request precise changes without rebuilding a project.
- **Transcript editing:** speech-to-text becomes an editable representation of the media; text operations can produce corresponding media edits.
- **AI co-editor / agent:** the AI can inspect the project, choose tools, execute multi-step operations, and report what it changed.
- **Traditional timeline:** multi-track editing, trimming, splitting, arranging, effects, transitions, text, audio, keyframes, layers, and fine timing control.
- **Visual intelligence:** search and organize footage by people, objects, environments, speech, metadata, scenes, and other visual signals.
- **Social repurposing:** automatic reframing and output adaptation for vertical, landscape, square, and other destinations.
- **Audio intelligence:** noise cleanup, voice enhancement, silence/filler removal, music mixing, ducking, beat-aware editing, and room-tone handling.
- **Creative compositing:** masks, background removal/replacement, overlays, tracking, effects, animation, and visual restyling.
- **Generative editing:** extend footage, generate supporting media or sound effects where available, and integrate generated assets directly into an editable sequence.
- **Collaboration:** shared projects, review/comment workflows, approvals, and team-oriented editing.
- **Publishing workflow:** captions, translation, metadata, platform-oriented exports, and multi-output generation.

3. Weaknesses / Loopholes / Improvement Opportunities

These are not claims that every product is defective. They are areas where the reference products leave room for a stronger editor, or where their current feature boundaries create friction.

OpusClip / AI Producer

Strength: Strong agentic first-draft editing, plain-language control, captions/B-roll/motion/music/SFX, edit logs, and a hands-on last-mile editor.

Improvement opportunities:

- Optimization is strongly associated with talking-head and short-form workflows; a broader story-first editor would cover more production types.
- AI decisions can still be constrained by the type of footage and workflow the product is optimized for.
- A stronger professional finishing layer would reduce the need to move complex work elsewhere.
- Creator-style automatic decisions need stronger project-wide continuity, narrative structure, and long-form control.
- AI should expose not only what changed, but dependencies, confidence, alternatives, and a reversible operation graph.

Descript

Strength: Transcript-first editing, Underlord AI co-editor, audio cleanup, filler/silence tools, regeneration, captions, translation, layouts, and collaboration.

Improvement opportunities:

- Text is an excellent control surface for speech-heavy content but cannot fully represent visual editing intent.
- Professional visual workflows benefit from deeper timeline, compositing, color, motion, and media-management controls.
- Transcript, generated content, and visual edits need stronger synchronization and provenance guarantees as AI edits become more complex.
- AI should be able to reason from the whole visual timeline, not only transcript semantics.
- A stronger unified project model should let text, scene, timeline, and AI operations modify the same underlying state.

CapCut

Strength: Large mobile/desktop creator toolkit, Auto Cut, Smart Search, Auto Reframe, captions, templates, effects, and broad AI creation.

Improvement opportunities:

- Feature breadth can create workflow complexity and inconsistency across platforms/regions.
- AI availability can vary by platform or region; the architecture should degrade cleanly and explain capability availability.
- Professional project organization, deterministic AI behavior, versioning, and advanced media management can be deeper.
- Creators need more explicit control over why AI selected shots, beats, transitions, or templates.
- A strong professional interchange/versioning layer would help users move from quick creator edits to serious projects.

Videoleap

Strength: Mobile-first creative effects, visual manipulation, compositing, and AI-assisted transformations.

Improvement opportunities:

- Creative effects are valuable but can be separated from deeper story planning and production management.
- Long-form project organization, advanced audio, professional color, and complex sequencing can be stronger.
- AI needs stronger project-level awareness so effects serve a coherent story rather than being isolated transformations.
- Cross-device project continuity and robust version history are important for larger projects.
- A unified AI planner could turn creative instructions into a sequence of coordinated operations.

Adobe Premiere

Strength: Professional timeline, media intelligence, AI Assistant, object masking, Generative Extend, generative media, color and workflow depth.

Improvement opportunities:

- Professional power produces a steeper learning curve; more tasks should be accessible through natural-language intent.
- AI assistance is increasingly capable but remains constrained by supported tools and beta/feature boundaries.

- Generative features can have concrete workflow limitations, such as unsupported transcription/media-intelligence/speed workflows for some generated extensions.
- A more universal operation graph could make AI changes easier to audit, undo, compare, and reproduce.
- Cloud-dependent AI should have transparent local/cloud routing and predictable fallback behavior.

4. PHENOVA Competitive Expansion — New Requirements

The following capabilities are added to PHENOVA as product requirements. They extend the existing specification rather than weakening or replacing it.

4.1 AI Director / Editing Agent

- Add an AI Director capable of planning and executing multi-step editing tasks from natural-language instructions.
- AI Director must inspect the actual project, media analysis, timeline state, capabilities, constraints, and user intent before acting.
- AI Director must be able to create a complete first cut, revise it, explain decisions, and apply targeted changes without unnecessarily rebuilding unrelated portions.
- Maintain an operation graph / edit log: every AI change records the affected assets, operations, reason, confidence where available, and rollback relationship.
- Support user modes such as automatic, guided, and manual control without creating separate project representations.

4.2 Story and Creative Intelligence

- Analyze project-level narrative structure: hook, setup, development, climax/primary point, resolution/call-to-action where applicable.
- Evaluate pacing, repetition, continuity, visual variety, speaker balance, shot quality, audio continuity, and unnecessary gaps.
- Allow creative goals such as cinematic, energetic, documentary, educational, promotional, vlog, interview, social short, or user-defined styles to influence the Edit Plan.
- Provide alternative edit proposals when multiple reasonable interpretations exist rather than silently committing to one.
- Distinguish objective corrections from subjective creative choices.

4.3 Unified Media Understanding

- Index speech/transcript, faces, speakers, objects, scenes, shots, camera motion, silence, audio characteristics, visual quality, metadata, and timeline context when supported.
- Support natural-language search such as finding a person, object, topic, reaction, location, phrase, or visual event.
- Expose only analysis that was actually performed; unsupported analysis must not be represented as fact.
- Cache analysis so repeated AI operations do not unnecessarily repeat expensive processing.

4.4 Dual Editing Surface

- Provide a conventional timeline and a transcript/document surface over the same underlying project state.
- Edits from either surface must remain synchronized.
- AI actions must modify the same canonical project representation used by manual editing.
- Users must be able to move from AI automation to frame-level/manual precision at any time.

4.5 Professional Editing Depth

- Expand the capability registry over time to include multi-track timeline operations, keyframes, masks, motion tracking, blend modes, advanced transitions, advanced color controls, professional audio processing, and more sophisticated compositing where technically supportable.
- Every new operation must be real, versioned, tested, and exposed only when supported by the active capability registry.
- Add project organization features for bins/folders, tags, labels, markers, searchable metadata, and large-project media management.

4.6 Audio Intelligence

- Add speech enhancement, noise/echo cleanup, silence detection, filler removal, room-tone-aware gap handling, volume normalization, ducking, beat analysis, and intelligent music placement where technically feasible.
- Preserve dialogue intelligibility and avoid destructive audio decisions without an auditable project operation.
- Allow the AI to reason about audio continuity when selecting cuts.

4.7 Smart Reframing and Multi-Output

- Automatically reframe subjects for multiple aspect ratios while preserving important visual information.
- Generate platform-specific variants from one canonical project without duplicating unrelated editing logic.
- Support user-defined safe areas, subject priority, caption regions, and brand layout constraints.

4.8 Creative Effects and Compositing

- Add background removal/replacement, object masking, tracking, overlays, controlled restyling, motion graphics, and effect stacks as real capabilities when implemented.
- AI should select effects according to the creative brief rather than applying effects merely because they are available.
- Maintain editable parameters and allow the user to remove or modify AI-selected effects individually.

4.9 Generative Editing Integration

- Architect a provider-neutral generative media interface for future integration of external video/image/audio generation providers.
- Generated assets must enter the project as identifiable, editable assets with provenance and clear distinction from original footage.
- Support generative extension/repair workflows where a compatible provider exists, while preserving the original media.
- Do not make Creative Generation a current dependency; the interface is an architectural provision for future expansion.

4.10 Collaboration and Review

- Add optional project sharing, comments, review links, approval states, and collaborator roles as a future capability.
- Preserve immutable project versions so collaboration does not destroy prior work.
- Support review notes that can be converted into structured AI correction requests.

4.11 AI Transparency and Control

- Provide an AI edit log showing what changed.
- Allow accept/reject of proposed changes where the workflow calls for review.
- Allow targeted undo/restore instead of only global undo.
- Show capability limitations before execution where possible.
- Never hide an AI failure behind a successful-looking render.
- For generated content, retain provenance and provider/model metadata internally where available.

4.12 Reliability and Fallback

- Extend the existing provider architecture so AI, media analysis, rendering, and future generation providers can be swapped without rewriting the project model.
- When a provider fails, retry or fall back according to capability and job policy rather than silently producing a degraded result.
- A project must remain editable if an optional AI provider becomes unavailable.
- Long-running AI/media tasks must use the existing job-state, progress, retry, cancellation, and output-validation architecture.

5. Competitive Feature Matrix — What PHENOVA Should Cover

Capability	OpusClip	Descript	CapCut	Videoleap	Premiere	PHENOVA addition
AI first-cut	Strong	Strong	Strong	Partial	Emerging	Required
Natural-language editing	Strong	Strong	Growing	Growing	Strong	Required
Transcript editing	Strong	Core	Strong	Partial	Strong	Add unified surface
Professional timeline	Moderate	Growing	Strong	Moderate	Very strong	Expand progressively
Visual/media intelligence	Strong	Growing	Strong	Moderate	Very strong	Required
Auto reframing	Strong	Strong	Strong	Strong	Strong	Required
Creative effects	Strong	Moderate	Very strong	Very strong	Very strong	Expand registry
Advanced audio	Moderate	Strong	Strong	Moderate	Very strong	Expand
Generative editing	Growing	Strong	Strong	Strong	Very strong	Provider-neutral future layer
Versioning / auditability	Opportunity	Moderate	Opportunity	Opportunity	Strong	Make foundational
AI explainability	Strong	Growing	Limited/variable	Variable	Growing	Make explicit
Project-wide creative reasoning	Opportunity	Opportunity	Opportunity	Opportunity	Emerging	Core differentiator

6. Changes to the Existing PHENOVA Architecture

- **Edit Plan:** expand the operation vocabulary and allow grouped/conditional operations while retaining strict schema validation, capability-registry validation, entitlement validation, and deterministic execution.
- **Media Understanding:** expand from duration/dimensions/FPS/codec/audio metadata and basic scene/silence/audio analysis to a cached multimodal project index.
- **AI Orchestration:** add a planning stage that can select analysis tools, editing operations, and optional generation providers before producing the executable Edit Plan.
- **Project State:** add a canonical timeline/scene/transcript representation so AI, transcript editing, manual editing, and corrections operate on the same project state.
- **AI Audit:** add AI operation logs and reversible operation groups linked to project versions.
- **Capability Registry:** add feature maturity, provider availability, platform support, resource requirements, and test coverage to each capability.
- **Render System:** preserve the existing queue/state/idempotency architecture while adding more advanced operation graphs and multi-output rendering.
- **Offline/Degraded:** keep optional AI features gracefully unavailable when network/provider access is missing; do not fabricate analysis or edits.
- **IFEC:** keep IFEC as an enhancement layer. New capabilities must use provider interfaces so PHENOVA remains independently functional.

7. New Testing Requirements

- AI first-cut tests using real media with known expected structural properties.
- Natural-language correction tests where only the requested project region changes and unrelated edits remain stable.
- Transcript/timeline synchronization tests in both directions.
- Media-intelligence retrieval tests for people, objects, speech, scenes, and metadata where each capability is implemented.
- Pacing/continuity tests covering dead air, repeated shots, abrupt audio changes, and invalid clip selections.
- Multi-aspect-ratio reframing tests verifying subject preservation and output dimensions.
- AI edit-log tests verifying every accepted AI operation is traceable and reversible.
- Provider failure tests for every new AI/media/generation provider interface.
- Generated-asset provenance tests where generative capabilities are later introduced.
- Large-project performance tests covering storage, indexing, timeline responsiveness, memory, thermal behavior, render concurrency, and recovery.

- Regression tests ensuring newly added capabilities never violate the existing no-fake-output, quota, security, ownership, persistence, or Definition-of-Done rules.

8. Updated Implementation Priority

- **Keep current phases 0–7 unchanged as the production foundation.**
- **Competitive Phase A:** strengthen media understanding and the canonical project model.
- **Competitive Phase B:** add AI Director / first-cut agent and operation log.
- **Competitive Phase C:** add transcript/document editing synchronized with the timeline.
- **Competitive Phase D:** expand professional timeline, audio, color, compositing, tracking, and creative capability registry.
- **Competitive Phase E:** add smart reframing and multi-output generation.
- **Competitive Phase F:** add optional collaboration/review.
- **Competitive Phase G:** introduce provider-neutral generative editing interfaces when the planned Creative Generation capability becomes active.
- Each phase must obey the existing Definition of Done: implementation + UI + backend integration + persistence + error handling + security + tests + real execution verification.

9. Updated Product Principle

PHENOVA should not merely accumulate features from five editors. The integrated requirement is to combine their strongest interaction models into one canonical project system: AI agent + transcript + timeline + creative tools + professional controls + media intelligence + reversible project history.

The key architectural rule is that these are not five separate editors inside one app. They are multiple interfaces over the same project state and the same validated Edit Plan / operation system.

10. Binding Compatibility With the Existing Specification

- The current PHENOVA identity, Android-first/mobile-first architecture, independent operation, and automatic IFEC enhancement remain unchanged.
- The current real-media requirement remains unchanged.
- The current strict Edit Plan validation remains unchanged and is expanded rather than weakened.
- The current quotas, Premium limits, billing rules, authentication, storage, security, offline behavior, testing, deployment, and acceptance requirements remain binding.
- No competitive feature is considered complete merely because a button, screen, API, JSON field, or model call exists.
- If a new capability cannot be implemented genuinely, it must be marked incomplete and must not be presented as working.

11. Research Basis and Current-Product Notes

Current official documentation confirms that OpusClip now positions AI Producer as an AI video editor agent that creates a first draft and supports plain-language changes, while retaining timeline/text-based editing. ■cite■turn0search14■turn0search16■

Descript currently combines transcript editing, a conversational AI co-editor called Underlord, audio cleanup, captions, translation, collaboration, and a traditional timeline/scene editor. ■cite■turn0search4■turn0search11■

CapCut documents Auto Cut for beat/speech/text-aware automatic cutting on mobile and desktop, plus Smart Search and Auto Reframe in its desktop AI editor. Some AI features vary by platform or region. ■cite■turn0search7■turn0search12■turn0search18■

Adobe Premiere currently documents AI Assistant for multi-step project/timeline work, Media Intelligence, Object Mask, Generative Extend, multilingual captions, and a Generative Media Tool. Adobe also documents concrete limitations for some Generative Extend workflows, including transcription/media-intelligence and speed-adjustment constraints.

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The supplied current PHENOVA specification already requires real editing, validated Edit Plans, media understanding, versioning, corrections, real rendering, offline/degraded behavior, IFEC fallback, security, automated testing, and a real end-to-end acceptance flow. This expansion therefore adds competitive capabilities without changing the foundational architecture rules. ■filecite■turn1file0■L4-L13■